Baseline A/A driver calibration

UTC run id: `run-20261011T1700Z`. Native run and comparison: completed, exit status 0.

All 72 requested A/A records completed. The formatted final driver, linked to the saved unoptimized baseline library, reproduced every baseline output, metadata field, allocation count and byte count exactly. The comparison’s optimization acceptance flag is false because identical baseline copies cannot satisfy improvement thresholds; calibration passed.

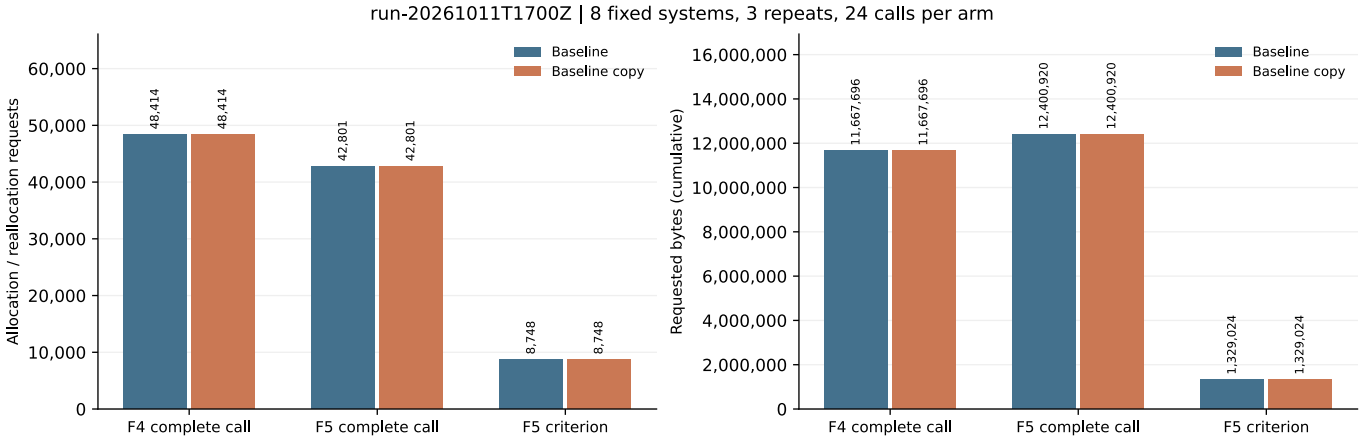


Figure 1: run-20261011T1700Z | 8 fixed systems, 3 repeats, 24 calls per arm. Recorded counts and cumulative requested bytes; 24 calls per arm.

Arm	Baseline requests	Other requests	Baseline bytes	Other bytes
criterion	8748	8748	1329024	1329024
f4	48414	48414	11667696	11667696
f5	42801	42801	12400920	12400920

Production baseline e957e48c2b8cf1f84a7fc44a1d673103964a789d; formatted diagnostic in ee344374. Original and formatted binaries were retained under /workspace/scratch. Hashes: records/baseline-aa-sha256.txt. Raw values and control: records/baseline.jsonl, records/baseline-aa.jsonl, records/baseline-aa-comparison.json.

The driver was compiled natively using rustc –edition=2021, -C opt-level=3, the saved baseline rlib and the original release serde_json, rayon and blake3 dependencies. Full reconstruction command is in the study report. Formatting did not alter the allocation accounting interval. No unsuccessful A/A records occurred.

Host and scope. Linux x86-64, AMD EPYC 9V74, 5 logical CPUs, 18,882,699,264 bytes RAM; Rust 1.99.0 (b940084d7), release, one Rayon worker. Exclusive CPU isolation was not established. Wall time and peak memory were not measured. This is a standalone Boolean matrix allocation diagnostic; full ECDLP cost, matched rho ratios, m=83 scaling and general-field gains remain unestablished.

Inputs and accounting. Eight planted quadratic systems: (n,m,d) = (8,8,3), (8,8,4), (10,10,4), (10,10,5), seeds `0x2545f4914f6cdd1d` and `0x9e3779b97f4a7c15`. Warmup precedes three repetitions. Each bar covers 24 calls. Fixture generation, hashing and verification are outside the audit; packing, elimination and output construction are included. Bytes are cumulative requests, not peak memory. Raw repeated values are retained; no statistical interval was estimated.

Evidence. Repository `aburan28/crypto`; study `research/f4_f5_row_build_20261011`. Native driver: `examples/f4_f5_row_build_bench.rs`. Full commands and paths: this directory’s `REPORT.md`; graph values: `graph-data.json`; rendering source: `../../render_reports.py`; raw records and hashes: `../../records/`.